



Pokhara University
Everest Engineering College
Final Assessment
Semester - Fall

Level: Bachelor	Year: 2025
Program: BE IT, 1st Semester	F.M: 100
Faculty: Science and Technology	P.M: 45
Subject: Calculus I	Time: 3 hrs.

*Attempt **all** the questions.*

1. (a) Find y_n if $y = \sin x$. State Leibnitz's Theorem. If $y = (\sin^{-1}x)^2$, then prove that
 - (i) $(1 - x^2)y_2 - xy_1 - 2 = 0$
 - (ii) $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$. Also, show that $y_{n+2}(0) = n^2y_n(0)$. [8]
- (b) State Lagrange's Mean Value Theorem and write its geometrical meaning. When it coincide with Rolle's Theroem? For what value of a, m, b does the function

$$f(x) = \begin{cases} 3 & \text{if } x = 0 \\ -x^2 + 3x + a & \text{if } 0 < x < 1 \\ mx + b & \text{if } 1 \leq x \leq 2 \end{cases}$$

satisfies the hypothesis of mean value theorem on $[0,2]$.

[7]

(OR)

Assuming the validity of the expansion, prove that

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

by using Maclaurin's series. Also, show that the function $f(x) = (x)^{\frac{5}{2}}$ **cannot** be expanded in Maclaurin's Series.

2. (a) Define asymptote to the curve. Find the asymptotes to the curve $x^3 + 3x^2y - 4y^3 - x + y + 3 = 0$. [8]
- (b) Show that the radius of curvature for the polar curve $r^m = a^m \cos m\theta$ at (r, θ) is $\rho = \frac{a^m}{r^{m-1}(m+1)}$. [7]
3. Solve the following integrals(**any three**):
 - (a) $\int \frac{x \tan^{-1} x}{(1+x^2)^{3/2}} dx$
 - (b) $\int_0^{\pi/2} \log(\sin x) dx = \int_0^{\pi/2} \log(\cos x) dx = \frac{\pi}{2} \log(\frac{1}{2})$
 - (c) $\int_1^{\infty} \frac{\log x}{x^2} dx$
 - (d) $\int_0^{2a} x^{9/2} \sqrt{2a-x} dx = \frac{21\pi a^6}{16}$. by using gamma function. [3 × 5 = 15]
4. (a) Find the volume of the solid generated by revolving the region enclosed by triangle with vertices (1,0), (2,1) and (1,1) about y-axis. [7]
- (b) State and prove the Euler's Theorem for two independent variables with degree n . If $\cos u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$, show that $x u_x + y u_y = \frac{1}{2} \cot u$. [8]
5. (a) If the sum of the dimensions of a rectangular swimming pool is a . Prove that the amount of water in the pool is maximum when it is a cube. [8]

- (b) Define Bernoulli's differential equation. Solve the differential equation $\frac{dy}{dx} + y \tan x = y^3 \sec x$. [7]

(OR)

The growth rate of a culture of bacteria is directly proportional to the number of bacteria present. After one day it is 1.5 times of original number. Find after how many days it will be (i) double (ii) triple.

6. (a) Define Euler-Cauchy equation. Solve the initial value problem

$$(4x^2 D^2 - 12x D + 16)y = 0, \quad y(1) = 0, \quad y'(1) = 3. \quad [7]$$

- (b) Find the general solution of $y'' + y = \sin x$. [8]

7. Attempt all the questions

- (a) Show that the function $f(x)$ defined by

$$f(x) = \begin{cases} x^2 + 1 & \text{if } x < 1 \\ 3x + 1 & \text{if } x \geq 1 \end{cases}.$$

is **not** differentiable at $x = 1$.

- (b) Discuss the symmetry of the curve $x^3 + y^3 = 3xy$. Does the curve pass through the origin? If yes, find the tangent to the curve at origin.
- (c) Show that the perimeter of the circle $x^2 + y^2 = a^2$ is $2\pi a$.
- (d) Solve: $\frac{dy}{dx} + \frac{1+\cos 2y}{1-\cos 2x} = 0$

$$[4 \times 2.5 = 10]$$